

EC-481: Wireless and Mobile Networks



Download Slides

Student Questions

Research and Teaching Experience

Highlights

- Assistant Professor, DCE, HITEC University, Taxila.
- Postgraduate Incharge, FYDP Incharge HITEC University.
- Member Board of PGS, CPED, UET Taxila.
- Total 13 years of Experience
- 11 Years Teaching Experience
- 2 Years Industry Experience
- Lead NCCS Lab
 - ✓ DPI LAB
- Research Grants from Pakistan (1), New Zealand (2).
- Fully Funded Scholarships for MS and PhD
- IF Journals: 06, Conferences(08)
- Accumulative IF>30
- Supervision : In progress (03 PhD, 02 MS, 02 BS)
- Completed: 05 MS, 3 BS.

HITEC University Taxila



- International Collaborations:
 - ✓ University of West London, UK
 - ✓ Technology Innovation Research Group, School of Information Technology, Whitecliffe, NZ.
 - ✓ Munster Technological University, Rossa Avenue, Bishopstown, Cork, Ireland.
- National Collaborations:
 - ✓ UET Taxila.
 - ✓ UoS, Sargodha.
 - ✓ Kohsar University Murree.
 - ✓ Bahria University Islamabad.

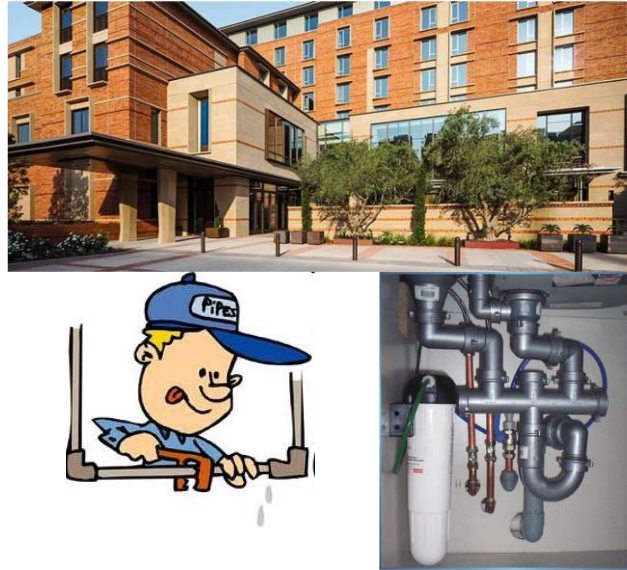


- ☐ Goal of this Course
- ☐ Grading
- ☐ Contents of the course
- ☐ Tentative Schedule

Student Questions

Networking = “Plumbing”

- ❑ Networking is the “plumbing” of computing
- ❑ Almost all areas of computing are network-based.
 - Distributed computing
 - Big Data
 - Cloud Computing
 - Internet of Things
 - Smart Cities



Networking is the backbone of computing

Student Questions

Networking is Fueling All Sectors of Economy

- ❑ Networking companies are among the most valued companies: Apple, AT&T, Samsung, Verizon, Microsoft, China Mobile, Alphabet, Comcast, NTT, IBM, Intel, Cisco, Amazon, Facebook, ...

⇒ All tech companies that are hiring currently are networking companies

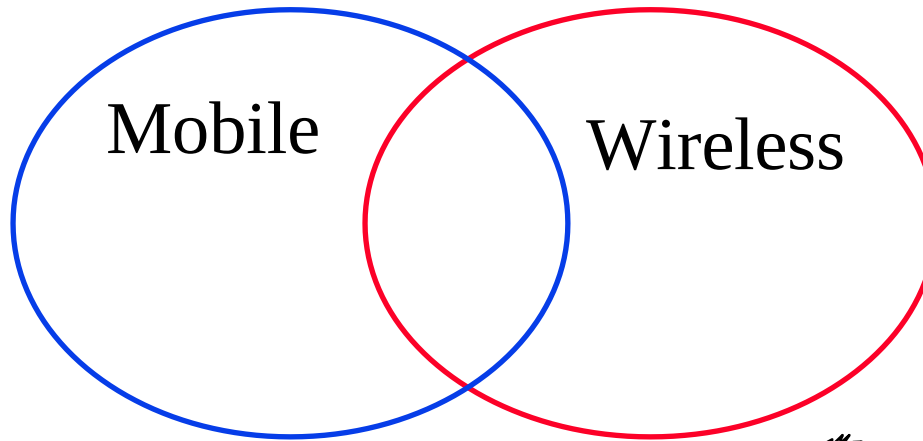
- ❑ Note: Apple became highly valued only after it switched from computing to communications (iPhone)



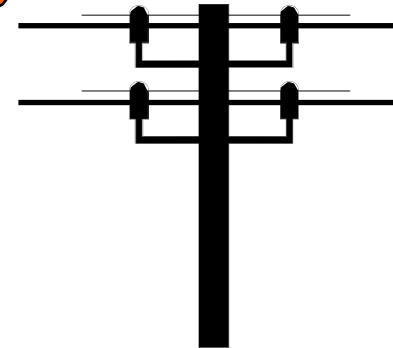
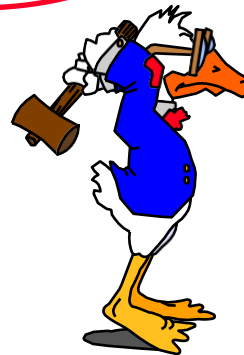
Networking = Economic Indicator

Student Questions

Mobile vs Wireless



- ❑ Mobile vs. Stationary
- ❑ Wireless vs Wired
- ❑ Wireless $\Rightarrow$ Media sharing issues
- ❑ Mobile $\Rightarrow$ Routing, addressing issues



Student Questions

Wireless Networking

Impact of Wireless on Networking:

1. Not tied to walls/infrastructure
⇒ Ad-hoc networking
2. Error-prone ⇒ Traffic Management
3. Frequent Disconnections
⇒ Resource Management
Quality of Service for multimedia
4. Battery operated
⇒ Media access and networking while sleep
⇒ Time synchronization
5. Broadcast ⇒ Security

Student Questions

Mobile Networking

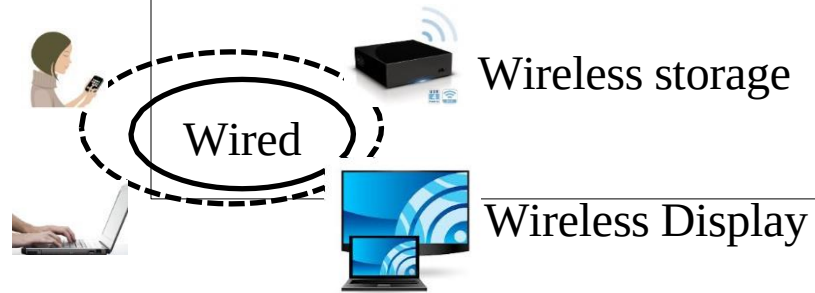
Impact of Mobility on Networking:

- ☐ Location
- ☐ Addressing
- ☐ Handoff

Student Questions

Why Wireless Networking?

1. Wireless (Wi-Fi) is ubiquitous (Intel Centrino)
2. Most of the access (end user connectivity) is wireless
 - Smart phones, Tablets, and many laptops (Ultra books) have no wired Ethernet connections
3. Most of telecommunication carriers' revenue is in wireless
4. New Developments:
 - 5G: 1 Gbps Metropolitan Area Networks
 - Vehicular Networking (802.11p)



Student Questions

Mobile Internet

- ❑ June 29, 2007: Apple announced iPhone
⇒ Birth of Mobile Internet, Mobile Apps
 - Almost all services are mobile apps: Google, Facebook, Bank of America, ...
- ❑ 2014 **mobile** data traffic was 2.5×10^{18} B/month.
30× the size of the entire global Internet in 2000 (75 PB/month).
- ❑ Between 2016-21:
 - PC traffic will be only $1/4^{\text{th}}$ compared to $1/2$ in 2016.
 - **Smart phone** traffic will be $1/3^{\text{rd}}$ compared to $1/8^{\text{th}}$ in 2016
 - **Mobile traffic** will grow twice as fast as fixed IP traffic
- ❑ Issues: Errors, Disconnection, Limited bandwidth, Limited distance



Student Questions

Ref: Cisco, "Cisco Visual Networking Index: Forecast and Methodology, 2016-2021," June 6, 2017,

<https://www.cisco.com/c/dam/en/us/solutions/collateral/service-provider/visual-networking-index-vni/complete-white-paper-c11-481360.pdf>

©2024 Sobia Arshad

Internet of Things



Smart Watch



Smart TV



Smart Car



Smart Health



Smart Home



Smart Kegs



Smart Space



Smart Industries

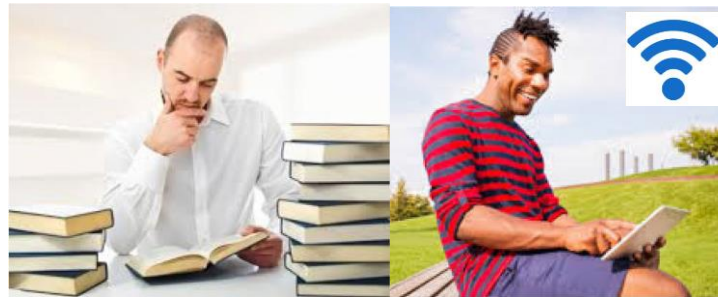


Smart Cities

Student Questions

What's Smart?

- ❑ Old: Smart = Can think $\Rightarrow$ Computation
= Can Recall $\Rightarrow$ Storage
- ❑ Now: Smart = Can find quickly, Can Delegate
 $\Rightarrow$ Communicate = Networking
- ❑ Smart Grid, Smart Meters, Smart Cars, Smart homes, Smart Cities, Smart Factories, Smart Smoke Detectors, ...



Not-Smart

Smart

Student Questions

Goal of This Course

- ❑ Comprehensive course on wireless and mobile networking
- ❑ Broad coverage of current key areas
- ❑ Topics of interest to industry
- ❑ Intro to physical layer “Wireless Communication.”
- ❑ Emphasis on lower layers: Layers 2, 3
- ❑ Emphasize both present (Industry standards and products) and near future (Research)
- ❑ Graduate course: (Advanced Topics)
⇒ Less reliance on one textbook

Student Questions

What Will You Learn?

1. How is wireless different from wired communication?
2. How does Wi-Fi work?
 1. How is the speed of Wi-Fi increasing from 10 Mbps to 10 Gbps?
 2. What is the difference between a/b/g/n/ac/ad/...
3. How is Bluetooth different from Wi-Fi?
4. How is ZigBee different from Wi-Fi?
5. What are the protocols that are used in IoT?
6. Why do we need new protocols for IoT?
7. What is the fundamental difference between 1G/2G/3G/4G/5G
8. What new features came in with 4G?
9. What new techniques enabled 5G?
10. What about 6G? When and how?

Student Questions

Prerequisite: EC-332

- ❑ Protocol Layers: ISO/OSI reference model
- ❑ TCP/IP protocol stack
- ❑ LAN Addressing: Unicast vs. multicast, Local vs. Global
- ❑ Extended LANs: Hubs vs. Bridges vs. Routers vs. Switches
- ❑ IPv4 and IPv6 Address: Public vs. Private Addresses
- ❑ Subnets
- ❑ Address Resolution Protocol (ARP)
- ❑ TCP connection setup, Checksum (pseudo-header), Slow start
- ❑ TCP vs. UDP

Student Questions

Text Book

- ❑ No one book covers the the breadth of the material in this course
- ❑ There will be a reading list with each lecture.
The list may include some books, websites, and Wikipedia links
- Text book:
 1. Wireless and Mobile Networking, 2022, CRC Press, 1st Edition, Mahbub Hasan, 2022, ISBN:9780367487355.
- Reference books:
 1. Mobile and Wireless Networks, Khaldoun Al Agha, Guy Pujolle, Tara Ali-Yahiya, 1st Edition, Willey, 2016, Print ISBN:9781848217140 |Online ISBN:9781119007548.
 2. Updated Research papers from ACM/IEEE/SPRINGER/Elsevier etc.

Student Questions

Projects

- ❑ Hands-on project or a survey paper related to the topics of the course
- ❑ Most survey topics will be assigned.
You can suggest hands-on projects for approval.
- ❑ Average 6 Hrs./week/person on project + 9 Hrs./week/person on class
- ❑ Recent Developments: Last 2 to 4 years
⇒ Not in books
- ❑ Will be published on arxiv,
Better ones may be submitted to magazines or journals.

Student Questions

Examples of Projects

- ❑ Implementation of a Mobile Aerial Wireless Network
- ❑ A Survey of Bio-Inspired Wireless Communication
- ❑ Cloud RAN: Basics, Advances and Challenges
- ❑ A Survey of Distributed Radio Systems
- ❑ Energy Efficient Wireless Communication Survey
- ❑ Long Range Low Power (LRLP) Wireless Network
- ❑ LTE-A for Device to Device and Machine-to-Machine Comm
- ❑ M2M Communication Scheduling using LTE/LTE-A
- ❑ Emerging MIMO Technologies
- ❑ Network Coding for Wireless Applications: A review
- ❑ A Survey of Self-Organizing Networks

Ref: Raj Jain, “Wireless and Mobile Networking (Spring 2016),” <http://www.cse.wustl.edu/~jain/cse574-16/index.html>

Student Questions

Examples of Projects

- ❑ Voice over LTE: Status and Migration Trends
- ❑ A Survey of Software-Defined Wireless Networks
- ❑ Virtualization in Wireless Networks
- ❑ Energy Efficiency in Wireless Networking Protocols
- ❑ Wireless Power Transfer – Concepts and Applications
- ❑ Survey of Low Altitude Unmanned Aerial Vehicles
- ❑ Security and Privacy Issues in the Internet of Things
- ❑ Wireless Networks for Disaster Relief
- ❑ Survey of Wireless Based Indoor Localization Technologies
- ❑ Recent Advances in Broadband Wireless Access Networks
- ❑ Recent Advances in Cognitive Radios
- ❑ Constrained Application Protocol for Internet of Things

Student Questions

Project Requirements

- ❑ Recent Developments: Last 3 to 4 years
 - ↳ Generally not in books
 - ❑ Comprehensive Survey:
Technical Papers, Industry Standards, Products
 - ❑ Will be published on arxiv website,
Better ones may be submitted to magazines or journals
 - ❑ No copyright violations:
 - ⇒ You need to re-draw all figures
 - ⇒ You need to summarize all ideas in your ***own*** words
 - ⇒ Cannot copy any part of text or figure unmodified
 - ⇒ Short quotes ok
 - ⇒ Any unmodified figures need permissions
- Any infringement will result in forfeiture of grades even after graduation.

Student Questions

Grading

- ❑ CH= 3 (3+0)
- ❑ Final = 50%
- ❑ Mid Term = 30%
- ❑ Assignments = 10%
- ❑ Quizzes = 10%

⇒ ❑ **Pass/Fail**: 60 points

⇒ ❑ Project is mandatory to pass course and question will be covered in Mid and Final Exam

Project Schedule and Point Weights

Due	Points
Search	0.5
HTML Sample	0.5
Topic	
References	1
Outline	1
Draft Paper	4
Reviews	1
HTML	2
Final Paper	5
Total	15

Student Questions

Questions

- ⇒ ☐ During the class, I may ask some questions to ensure that you have understood the topics.

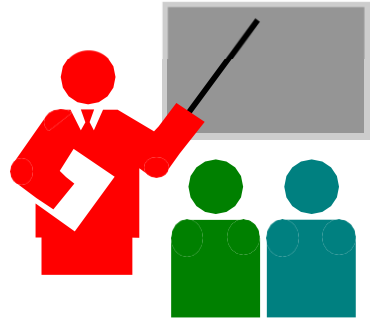
Student Questions

Academic Integrity

- ❑ Academic integrity is expected in homework, quizzes, and exams.
- ❑ All solutions submitted are expected to be yours and not copied from others or solution manuals, or from the Internet
- ❑ University requires us to report all integrity violations to the department

Student Questions

Summary



- ❑ We are living in the Internet age. Most activities, including work and play, require the Internet.
- ❑ Networking companies are among the most valued companies in the world.
- ❑ Wireless networking is taking over the edge, fueled by smartphones and smart devices (IoT).
- ❑ Goals of this course:
 - To prepare you for the current job market in networking
 - To teach you how to keep up with the latest in wireless and mobile networking

Student Questions

Google Search Modifiers

- ☐ filetype:pdf, doc, ppt, pptx
- ☐ site:google.com
- ☐ intitle:trend
- ☐ inurl:trend
- ☐ allintitle:Networking Trends
- ☐ Allinurl:
- ☐ “ “ $\Rightarrow$ Exact Phrase
- ☐ OR
- ☐ AND
- ☐ + $\Rightarrow$ Must include
- ☐ - $\Rightarrow$ Not include
- ☐ ~X $\Rightarrow$ X or similar
- ☐ * $\Rightarrow$ Wildcard

Student Questions

Project Homework 1

- ❑ Search web pages, books, and journal articles from IEEE Xplore, ACM Digital Library, **MOBIUS**, **Safari books**, **ILLIAD** at Olin Library for one of the following topics:
 1. Wireless Networking Trends
 2. Mobile Networking Trends
 3. Internet of Things
- ❑ On the web, try the following search points:
 - <http://scholar.google.com>
 - <http://books.google.com>
 - <http://dl.acm.org/>
 - <http://ieeexplore.ieee.org/Xplore/home.jsp>

Student Questions

Project Homework 1 (Cont)

- ⇒ ☐ Ignore all entries dated 2019 or before. Also ignore all entries that do not indicate the topic or similar words in the title. List others in the following format (5 each):
 - Author, “Title,” publisher, year, ISBN. (for five books)
 - “Title,” URL [One line description] (for five webpages)
 - Author, “Title,” source (for five technical/magazine articles)
- ☐ For the books, including whether the book is available at Amazon, Safari, or ILLiad
- ☐ Serially number the references and submit them electronically to GCR.
- ☐ Make a list of other interesting search points and share them in class.

Student Questions

Common Mistakes in Project Homework #1

- ❑ Not indicating where the book can be found in google
- ❑ Listing books/Magazines/journals that have little to do with the topic – may show up in search engines because of a minor mention of the topic or words
- ❑ Web Pages – No one line descriptions
- ❑ Incomplete bibliographic data for journal articles. Need volume, issue, year, pages.
- ❑ Missing journals. Need names of journals dealing with the topic chosen.

Student Questions

Quiz 0: Prerequisites

True or False?

T F

1. ☐ ☐ Datalink refers to the 2nd layer in the ISO/OSI reference model
2. ☐ ☐ HTTP is an example of an application layer protocol
3. ☐ ☐ Finding a path from one node to another in a large network is a transport layer function.
4. ☐ ☐ CRC is used for error detection
5. ☐ ☐ MAC address of a node changes as it changes its location.
6. ☐ ☐ For long delay paths, on-off flow control is better than window flow control.
7. ☐ ☐ Ethernet uses a CSMA/CD access method.
8. ☐ ☐ All Ethernet packets are acknowledged.
9. ☐ ☐ The packets sent in a connection-oriented network are called datagrams.
10. ☐ ☐ Spanning tree algorithm finds a loop-free path in a network.

Marks = Correct Answers ____ - Incorrect Answers ____ = ____

Student Questions

Answers to this quiz
will be provided in the
class for you to check
your answers privately.

Acronyms

❑ BAN	Body Area Networks
❑ CSMA/CD	Carrier Sense Multiple Access with Collision Detection
❑ IEEE	Institution of Electrical and Electronic Engineers
❑ ILLIAD	Inter-Library Loan
❑ IMT	International Mobile Telecommunication
❑ IPv4	Internet Protocol Version 4
❑ IPv6	Internet Protocol Version 6
❑ ISO	International Standards Organization
❑ LAN	Local Area Network
❑ LRLP	Long Range Low Power
❑ LTE	Long-Term Evolution
❑ MAC	Media Access Control
❑ OSI	Open System Interconnection
❑ OSPF	Open Shortest Path First

Student Questions

Acronyms (Cont)

☐ RFID	Radio Frequency Identification
☐ TCP	Transmission Control Protocol
☐ TV	Television
☐ UMB	Ultra-Mobile Broadband
☐ URL	Uniform Resource Locator
☐ UWB	Ultra-Wideband
☐ VoIP	Voice over IP
☐ Wi-Fi	Wireless Fidelity
☐ WWW	World-Wide Web

Student Questions

Scan This to Download These Slides



<https://classroom.google.com/w/NjY0MjI0MjY2MzIz/t/all>

Student Questions